

Assignment 1: Understanding AI and Generative AI

Artificial Intelligence (AI) is a technology that enables computers to perform tasks that usually require human intelligence, such as understanding information, solving problems, recognizing patterns, and making decisions. Generative AI is a branch of AI that can create new content like text, images, lesson plans, worksheets, quizzes, and other learning resources based on the instructions given by the user.

In education, AI has become a valuable support tool for both teachers and students. As a Mathematics teacher, I find that AI can help identify students who need extra practice, monitor their learning progress, and suggest activities suited to different learning levels. Generative AI tools like ChatGPT and Google Gemini make lesson preparation much easier by helping create worksheets, competency-based questions, real-life word problems, and assessments in a short time.

Students can also use these tools to revise concepts, understand step-by-step solutions, and strengthen their problem-solving skills. However, AI cannot replace a teacher's subject knowledge, classroom experience, or professional judgement. Instead, it should be viewed as a teaching assistant that supports teachers in making mathematics learning more engaging, organized, and effective when used responsibly.

Assignment 2: AI and Education

Artificial Intelligence is becoming an important part of today's education system. It offers several benefits that make teaching and learning more effective. One of its greatest strengths is its ability to personalize learning. AI can provide practice questions and activities based on each student's learning level, helping them learn at their own pace.

For Mathematics teachers, AI reduces the time spent on routine tasks such as preparing worksheets, generating quizzes, designing competency-based questions, and organizing assessment records. This allows teachers to spend more time explaining concepts, improving students' problem-solving skills, and giving individual attention where needed.

At the same time, AI should be used carefully. Sometimes it may generate incorrect calculations, incomplete explanations, or examples that are not fully aligned with the NCERT textbook or CBSE curriculum. If students depend completely on AI without understanding the concepts or checking the answers, it may affect their logical reasoning and critical thinking skills. There are also concerns about data privacy, fairness, and unequal access to technology in different schools.

UNESCO emphasizes that AI should support teachers rather than replace them. Therefore, Mathematics teachers should always review and verify AI-generated content before using it in the classroom. With proper guidance, ethical use, and human supervision, AI can become a powerful educational tool that enhances mathematics teaching while maintaining accuracy, fairness, and meaningful learning.

Assignment 3: Reflection on Using Generative AI (Mathematics Teacher's Perspective)

While preparing a lesson on "**Exploring Algebraic Identities**" for Class IX, I used ChatGPT as a teaching support tool. It helped me prepare a structured lesson plan, frame clear learning objectives, and generate competency-based as well as higher-order thinking questions. It also suggested interesting classroom activities and real-life application-based problems, which made the lesson more engaging for students. Another advantage was that it provided differentiated practice questions suitable for learners with different abilities, saving me considerable preparation time.

However, while reviewing the material, I noticed that a few examples were not arranged according to the sequence followed in the NCERT textbook, and some questions needed to be modified to match the latest CBSE competency-based assessment pattern. Before using the content in class, I carefully verified the concepts using the NCERT textbook, CBSE guidelines, and my own teaching experience. I also edited a few activities and examples so that they were more suitable for the learning level of my students and the classroom environment.

This experience helped me realize that Generative AI is a useful teaching assistant rather than a replacement for a teacher. It can save time, provide creative ideas, and make lesson planning more efficient. At the same time, it is the responsibility of the teacher to verify every AI-generated resource to ensure that it is mathematically correct, aligned with the prescribed curriculum, and appropriate for students. When combined with a teacher's knowledge, experience, and professional judgement, AI can greatly enhance the teaching and learning of Mathematics.